Seongju Lee

Ph.D. in AI Convergence

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SUMMARY

Senior Researcher at HD Hyundai Robotics specializing in robust visual perception and multimodal learning. Research spans object detection, semantic segmentation, domain generalization, and multi-agent trajectory prediction. Experience leading perception software development and building multi-view datasets for autonomous systems. Current work focuses on humanoid and dual-arm robot learning and multimodal data pipelines.

SKILLS

Perception: Object Detection, Semantic Segmentation, Tracking
Learning: Domain Generalization, Knowledge Distillation, Multimodal Learning
Programming: Python, C++, C, MATLAB
Frameworks: PyTorch, ROS/ROS 2, LeRobot
Simulation: Isaac Sim, MuJoCo
Robot Learning: Imitation Learning, VLA, WAM
Robot Platforms: UR5, Franka Panda, HD Hyundai Robotics dual-arm robot
Languages: Korean (Native), English

RESEARCH EXPERIENCE

Jul. 2026 – Present **Senior Researcher** — **HD Hyundai Robotics**
Developing intelligent dual-arm robotic systems, multimodal data pipelines, and ROS 2 interfaces.

Mar. 2026 – Jun. 2026 **Postdoctoral Researcher** — **GIST Institute of Artificial Intelligence**

SELECTED PUBLICATIONS

Published Papers (*: Equal Contribution)

Journal quartiles: best JCR subject-category quartile (2025). Full publication list: Google Scholar.

Conference
MV2: A Large-Scale 360-degree Multi-View Maritime Vision Dataset for Object Detection and Segmentation
The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)
Junseok Lee*, Jongwon Kim*, **Seongju Lee*** (Co-first author), Taeri Kim, and Kyoobin Lee
[Paper] [Website]

Top Conference
MART: MultiscAle Relational Transformer Networks for Multi-agent Trajectory Prediction
The 18th European Conference on Computer Vision (ECCV 2024), Poster
TL;DR: Graph Neural Networks (GNNs), Hypergraph Transformer
Seongju Lee (First author), Junseok Lee, Yeonguk Yu, Taeri Kim, and Kyoobin Lee
[Paper] [Code] 27.9% Acceptance Rate Cited by 80

Top Conference
CD-FKD: Cross-Domain Feature Knowledge Distillation for Robust Single-Domain Generalization in Object Detection
2026 IEEE International Conference on Robotics and Automation (ICRA 2026)
TL;DR: Knowledge Distillation, Robust Object Detection, Single-Domain Generalization
Junseok Lee*, Sungho Shin*, **Seongju Lee**, Kyoobin Lee
[Paper] [Code] 38.04% Acceptance Rate Cited by 1

Selected Publications (continued)

Conference

Robust Maritime Object Detection under Adverse Conditions via Joint Semantic Learning without Extra Computational Overhead

The 2025 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS 2025)

Junseok Lee, **Seongju Lee**, Jongwon Kim, Jumi Park, and Kyoobin Lee

[Paper] 46% Acceptance Rate Cited by 2

Top Conference

Block Selection Method for Using Feature Norm in Out-of-Distribution Detection

2023 IEEE/CVF Computer Vision and Pattern Recognition Conference (CVPR 2023), Poster

Yeonguk Yu, Sungho Shin, **Seongju Lee**, Changhyun Jun, and Kyoobin Lee

[Paper] [Code] 25.8% Acceptance Rate Cited by 82

Top Conference

ManipForce: Force-Guided Policy Learning with Frequency-Aware Representation for Contact-Rich Manipulation

2026 IEEE International Conference on Robotics and Automation (ICRA 2026)

TL;DR: Imitation Learning, Multimodal Policy, Vision Foundation Models (VFMs), Force/Torque Sensor

Geonhyup Lee, Yeongjin Lee, Kangmin Kim, **Seongju Lee**, Sangjun Noh, Seunghyeok Back, Kyoobin Lee

[Paper] [Website] [Code] 38.04% Acceptance Rate Cited by 29

Journal

Human-efficient annotation system via active semi-supervised learning and selective review

Intelligent Systems with Applications, Vol. 31, Article 200701, 2026

Jongwon Kim, Youmin Park, Seunghyeok Hyun, Yunjae Heo, **Seongju Lee**, Yeonguk Yu, and Kyoobin Lee

[Paper] JCR 2025 Q1

Manuscript in Submission

In Submission

Rethinking Deformable Convolution as an Adapter with Cross-layer Weight Sharing for Robust Semantic Segmentation in the Wild

Target Journal: IEEE Robotics and Automation Letters

First Prize in the USV-based Obstacle Segmentation Challenge at the 3rd Maritime Computer Vision Workshop

TL;DR: Vision Foundation Models (VFMs), Domain Generalized Semantic Segmentation (DGSS), Deformable Convolution Networks (DCNs), Parameter-Efficient Fine-Tuning (PEFT)

Seongju Lee (First author), Raeyoung Kang, Sangbeom Lee, Sangjun Noh, Yeonguk Yu, Junseok Lee, and Kyoobin Lee

SELECTED PROJECTS

Apr. 2023 – Dec. 2024
Project Member
Software Dev. Leader

Development of a Situational Awareness System for Preventing Collisions and Accidents of Autonomous Ships

Ministry of Trade, Industry and Energy (MOTIE), South Korea

TL;DR: Multi-camera maritime object detection and tracking, training dataset creation, and integration into autonomous ship systems.

Apr. 2020 – Nov. 2024
Project Leader
AI & Software Dev. Leader

Development of Swarm USV Autonomous Navigation Algorithm

Agency for Defense Development (ADD), South Korea

TL;DR: Maritime object detection and intention prediction for swarm situational awareness, integrated into multiple USVs. [News]

Jul. 2026 – Present
AI & Software Developer

Development of an Intelligent Dual-Arm Robot

HD Hyundai Robotics

TL;DR: Robot learning with VLA/WAM models; VR/glove teleoperation and hand retargeting; simulation and multimodal demonstration data pipelines.

Sep. 2023 – Feb. 2026
Project Member
Software Developer

Development of a Micro Intelligent Drone for Autonomous Missions in Communication/GPS-denied Environments

Institute of Civil-Military Technology Cooperation (ICMTC), South Korea

TL;DR: RGB-IR object detection and a multimodal RGB, IR, and LiDAR database captured from drone views.

Selected Projects (continued)

Mar. 2022 – Dec. 2022
Project Member
AI & Software Dev. Leader

Ceiling Layout Estimation for Indoor Environments
LG, South Korea
TL;DR: Ceiling segmentation and dataset construction for robust floor plan generation under occlusion.

Apr. 2019 – Mar. 2020
Project Member
AI & Software Developer

Robotic IKEA Furniture Assembly from Assembly Manuals
Ministry of Science and ICT (MSIT), South Korea
TL;DR: Assembly-part instance segmentation and workflow visualization for robot assembly from IKEA manuals. [Video]

EDUCATION

Mar. 2020 – Feb. 2026
Gwangju Institute of Science and Technology (GIST) **Advisor: Prof. Kyoobin Lee**
Ph.D. in AI Convergence
Dissertation Title: Robust Perception and Trajectory Forecasting for Autonomous Navigation in the Wild

Mar. 2018 – Feb. 2020
Gwangju Institute of Science and Technology (GIST)
M.S., School of Integrated Technology (Intelligent Robotics Program)

Mar. 2014 – Feb. 2018
Gwangju Institute of Science and Technology (GIST)
B.S. in Mechanical Engineering

SELECTED AWARDS

Jun. 2026
Multimodal Semantic Segmentation Challenge, 4th Workshop on Maritime Computer Vision (MaCVi) **IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2026)**
First Place

Jan. 2026
RoCo Challenge @ AAAI 2026: Robotic Collaborative Assembling for Human-Centered Manufacturing **AAAI, A*STAR, NTU**
Finalist Excellence Award

Nov. 2025
Autonomous Driving AI Challenge (Semantic Segmentation) **MSIT, IITP, KADIF**
Second Prize (IITP President's Award)

Dec. 2024
USV-based Obstacle Segmentation Challenge, 3rd Maritime Computer Vision Workshop **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025)**
First Prize

Dec. 2024
USV-based Panoptic Segmentation Challenge, 3rd Maritime Computer Vision Workshop **IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2025)**
Second Prize